Given the road signs in figure 14.8, however, this is not possible, and we need to take detours: in the next sub-section, we'll see how.

14.3.2 Breadth First Search

Listing 14.2 shows the pseudo-code for Breadth First Search (*BFS*), whose goal, as the name suggests, is to widen the search as much as possible, keeping a perimeter of already visited vertices, and expanding this perimeter to the neighboring vertices: this is shown in figure 14.10, where the graph in figure 14.6.C is used to demonstrate the first few steps of this algorithm.

Listing 14.2 The bfs method

```
function bfs(graph, start,
isGoal)                                                    #1
  queue ← new
Queue()
#2

queue.insert(start)
             #3
  distances ← new
HashTable()                                                      #4
  parents ← new
HashTable()
#5
  for v in graph.vertices
do                                                         #6
    distances[v] ← inf
    parents[v] ← null
  distances[start] ←
0                                                                #7
```